Engineering Handbook – Volume 09

Textile Domain Model & Industrial Processes (Version 1.0)

Purpose

Documents the textile-processing knowledge that underpins the Industrial IoT platform, ensuring the software reflects real factory operations.

Factory Hierarchy

Company → Plant → Department → Machine → Component → Sensor → Telemetry. Textile-specific entities are implemented as configurable data rather than hardcoded logic.

Primary Assets

Jet Dyeing Machines, Stenters, Thermic Fluid Heater, Boilers, Air Compressors, Pumps, Blowers, Water Treatment Plant, Utility Meters and Laboratory equipment.

Jet Dyeing Process

Typical stages include loading, pretreatment, dosing, heating, dyeing, holding, cooling, rinsing and unloading. Machine telemetry should capture temperatures, pressures, motor currents, pump status and production information.

Stenter Process

Track chamber temperatures, fan motors, exhaust fans, conveyor speed, fabric width, moisture, gas consumption and production meters.

Utilities

Monitor electricity, steam, water, compressed air and thermic fluid. Utility data should be linked to production batches for efficiency analysis.

Recipe & Costing

Future OCR modules will digitize handwritten recipes, calculate chemical consumption, estimate production cost and maintain recipe history.

KPIs

Energy per kilogram, water per kilogram, steam consumption, machine utilization, downtime, production efficiency, alarm frequency and maintenance indicators.

Predictive Maintenance

Use vibration, current, temperature, runtime and alarm history to estimate component health and schedule maintenance.

Future Vision

Expand the same architecture to food, chemicals, pharmaceuticals and other process industries while preserving textile-specific extensions as modular packages.

Recommended Expansion

Future versions will include detailed process flow diagrams for dyeing, finishing, utility systems, laboratory workflows, quality control, inventory, production planning and AI-assisted recipe optimization.